

Permanent Mathematical Notes with a Deliberately Long Explanatory Title

Stable references for notes that continue to grow.

Permanent Mathematical Notes with a Deliberately Long Explanatory Title

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Suggested citation

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Chapter 1

The first tagged chapter (Tag 0AB)

1.1 A published section (Tag 0ABCD)

1.1.1 Published statements (Tag 0ABCDS1)

Theorem 1.1.1 (Fundamental comparison) (Tag 0ABCDEF). If two objects satisfy the same universal property, they are uniquely isomorphic in the appropriate sense.

Definition 1.1.2 (Tag 0ABCDE1). A published object is an entry whose semantic content and permanent tag have been recorded in the global registry.

Theorem 1.1.3 (Tag 0ABCDE2). A permanent tag does not require the statement to have a title.

Published local references stay compact: [Chapter 1](#) (Tag 0AB), [Section 1.1](#) (Tag 0ABCD), [Subsection 1.1.1](#) (Tag 0ABCDS1), [Theorem 1.1.1](#) (Tag 0ABCDEF), and [Definition 1.1.2](#) (Tag 0ABCDE1). The old identifier remains queryable as the frozen record [Tag System Test Book, Theorem 1.1.0, retired Tag 0ABCDE0; replaced by Tag 0ABCDEF].

1.1.2 Mathematical alphabets (Tag Θ ABCD $\mathcal{S}2$)

The inline specimen $\mathcal{M} \in \mathcal{C}$, $f : X \rightarrow Y$, $H^i(X, \mathcal{M}) \cong \mathbb{Q}^{\oplus r_i}$, and $\mathfrak{g} = \mathfrak{sl}_n(k)$ shows the selected families in running text. The full alphabet comparison is

Roman : ABCDEFGHIJKLMNOPQRSTUVWXYZ,
 abcdefghijklmnopqrstuvwxyz,
Italic : ABCDEFGHIJKLMNOPQRSTUVWXYZ,
 abcdefghijklmnopqrstuvwxyz,
Bold : ABCDEFGHIJKLMNOPQRSTUVWXYZ,
 abcdefghijklmnopqrstuvwxyz,
 Sans : ABCDEFGHIJKLMNOPQRSTUVWXYZ,
 abcdefghijklmnopqrstuvwxyz,
 Mono : ABCDEFGHIJKLMNOPQRSTUVWXYZ,
 abcdefghijklmnopqrstuvwxyz.

The decorative and semantic families are

Calligraphic : *ABCDEFGHIJKLMNOPQRSTUVWXYZ,*
 abcdefghijklmnopqrstuvwxyz,
 Script : *ABCDEFGHIJKLMNOPQRSTUVWXYZ,*
 abcdefghijklmnopqrstuvwxyz,
 Blackboard : ABCDEFGHIJKLMNOPQRSTUVWXYZ,
 abcdefghijklmnopqrstuvwxyz,
 Fraktur : **ABCDEFGHIJKLMNOPQRSTUVWXYZ,**
 abcdefghijklmnopqrstuvwxyz.

Chapter 2

A draft chapter (Draft tag 0CD)

2.1 Every theorem-style environment (Draft tag 0CDEF)

2.1.1 Statements, proofs, steps, and cases (Draft tag 0CDEFS1)

Slogan *Stable identity belongs to the mathematics, not to its current page number.*

Theorem A (The gallery theorem). Every supported statement style remains readable in long-form notes.

Proposition 2.1.1 (A moved statement) (Tag 0ABEF01). A mathematically unchanged published entry keeps its tag after it moves.

Theorem 2.1.2. This draft theorem has no provisional tag, so its reference contains only the current theorem number.

Lemma 2.1.3 (Local cancellation). If $g \circ f$ and g are isomorphisms, then f is an isomorphism.

Corollary 2.1.4. An identity morphism is an isomorphism.

Conjecture 2.1.5 (A plausible extension). The same comparison principle persists in the derived setting.

Question 2.1.6. Which hypotheses can be weakened without changing the conclusion?

Example 2.1.7 (Draft tag 0CDEF01). This valid-looking draft tag is visible but is not globally reserved.

Exercise 2.1.8 (Check the universal property). Prove the uniqueness assertion in [Theorem 1.1.1](#) (Tag 0ABCDEF).

Construction 2.1.9 (A fiber product). For maps $X \rightarrow S \leftarrow Y$, form $X \times_S Y$ by its universal property.

Notation 2.1.10. Write $D^b(X)$ for the bounded derived category of coherent sheaves on X .

Remark 2.1.11 (Draft tag 0CDEF01). Duplicate provisional tags are permitted in a draft scope and must be resolved before publication.

Proof sketch for Theorem A. The proof can be divided into referenceable steps and cases.

- Step 1 (Set up the comparison).** Choose the canonical maps supplied by the two universal properties.
- Claim 2.1.12 (The composites are identities).** Both composites agree with the corresponding identity morphisms.
- Case 1 (The source is initial).** Uniqueness of a map from the source proves the first identity.
- Case 2 (The target is terminal).** Uniqueness of a map to the target proves the second identity.

Thus [Step 1](#), [Claim 2.1.12](#), and [Case 1](#) complete the argument. □

Draft references omit their provisional tags: [Chapter 2](#), [Section 2.1](#), [Subsection 2.1.1](#), [Theorem 2.1.2](#), [Example 2.1.7](#), and [Remark 2.1.11](#).

2.2 Typesetting features (Draft tag 0CDGH)

An aligned equation can carry an ordinary label:

$$Lg^*Rf_*\mathcal{F} \simeq Rf'_*Lg'^*\mathcal{F}. \tag{2.1}$$

The reference [eq. \(2.1\)](#) uses the standard `cleveref` behavior.

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

Nested lists use alphabetic and roman labels:

- (a) Verify existence.
- (b) Verify uniqueness.
 - (i) compare the two candidates;
 - (ii) apply the universal property.

Scope	Tag length	Example
Book	1	0
Chapter	3	0AB
Section	5	0ABCD
Entry or subsection	7	0ABCDEF

```

1 def compose(f, g):
2     """Return the composite g after f."""
3     return lambda x: g(f(x))

```

Listing 2.1: A small source-code listing

Both [Code 2.1](#) and citations such as [[Har77](#)] are available without loading extra packages in the note source.

Bibliography

[Har77] Robin Hartshorne. *Algebraic Geometry*. Springer, 1977 (cit. on p. [8](#)).